Hierarchical Reinforcement Learning Framework for Two-Stage Stochastic Scheduling

1 Problem Formulation

We formulate the two-stage stochastic programming problem as a Hierarchical Markov Decision Process (MDP) spanning T discrete time steps.

1.1 Stage 1: Dynamic Pricing

- **State (S_t)**: $S_t = (\{l_{mt}, e_{mt}\}_{m=1}^{n}, \boldsymbol{\lambda_t}, \boldsymbol{p_t^F})$, capturing the locations l and energy levels e of n MCVs, background demand rates $\boldsymbol{\lambda_t}$, and fixed charging station prices $\boldsymbol{p_t^F}$.
- **Action ($\boldsymbol{p_t^M}$)**: The first-stage pricing decision generated by the macro-policy $\pi_{\theta_1}(\boldsymbol{p_t^M}|S_t)$.
- **Reward (R_t^p)**: Base reward (e.g., immediate revenue or penalty) realized at step t right after pricing.

1.2 Intermediate Phase: EV Choice Realization

After the price $\boldsymbol{p_t^M}$ is broadcasted, EV owners make their charging decisions, yielding the actual charging requests $\boldsymbol{D_t} = \mathcal{D}(\boldsymbol{\lambda_t}, \boldsymbol{p_t^M}, \boldsymbol{p_t^F})$.

1.3 Stage 2: MCV Multi-agent Dispatching

Let n be the number of Mobile Charging Vehicles (MCVs). The central dispatcher controls all n agents simultaneously.

- **State ($\tilde{S}_t$)**: $\tilde{S}_t = (\{l_t, e_t\}, \boldsymbol{p_t^M}, \boldsymbol{p_t^F}, \boldsymbol{D_t})$, including the actual charging requests $\boldsymbol{D_t}$.
- **Joint Action ($\mathbf{a}_t$)**: The discrete action for individual agents, defined as a_t. Each individual action $a_t \in \{I \cup J\} \times \{\text{service, idle, charge, discharge}\}$, where I_t represents the dynamic request nodes derived from $\boldsymbol{D_t}$.
- **Reward (R_t^d)**: The operational profit/cost realized from the dispatching at step t.

2 Neural Network Architecture

To handle the integer variables and avoid the action space explosion $O(|I_t \cup J|^n)$, we utilize a multi-agent policy network.

2.1 Stage 1 Networks (Pricing)

- **Actor** $\pi_{\theta_1}(\boldsymbol{p_t^M}|S_t)$: Outputs the distribution for the first-stage price $\boldsymbol{p_t^M}$.
- **Critic** $V_{\phi_1}(S_t)$: Estimates the expected total discounted return (pricing + dispatching) from step t onwards over all possible demand realizations and future transitions:

$$V_{\phi_1}(S_t) \approx \mathbb{E}_{\pi_1,\pi_2}\left[\sum_{\tau=t}^{T}\gamma^{\tau-t}(R_\tau^p + R_\tau^d)\middle| S_t\right] \tag{1}$$

2.2 Stage 2 Networks (MCV Dispatcher)

- **Actor** $\pi_{\theta_2}(\mathbf{a}_t|\tilde{S}_t)$: Output the distribution for the second-stage action.

  Invalid actions(nodes) are masked by setting their pre-Softmax logits to $-\infty$.
- **Critic** $V_{\phi_2}(\tilde{S}_t)$: Estimates the expected future return from the dispatch phase of step t to terminal step T:

$$V_{\phi_2}(\tilde{S}_t) \approx \mathbb{E}_{\pi_1,\pi_2}\left[R_t^d + \sum_{\tau=t+1}^{T}\gamma^{\tau-t}(R_\tau^p + R_\tau^d)\middle|\tilde{S}_t\right] \tag{2}$$

3 Interleaved MDP Formulation and Credit Assignment

To enable end-to-end joint training, each discrete time step $t \in \{1,\ldots,T\}$ is modeled as an interleaved sequence containing two sub-decision steps:

$$S_1 \xrightarrow{\boldsymbol{p_1^M}} (R_1^p, \tilde{S}_1) \xrightarrow{\mathbf{a}_1} (R_1^d, S_2) \xrightarrow{\boldsymbol{p_2^M}} \ldots \xrightarrow{\mathbf{a}_T} (R_T^d, S_{T+1}) \tag{3}$$

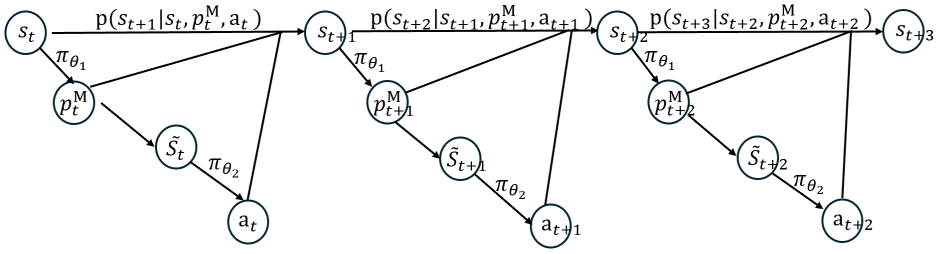


Figure 1: MDP Dynamics

This two-stage transition structure allows us to train both Pricing Policy π_{θ_1} and Dispatching Policy π_{θ_2} jointly via Proximal Policy Optimization (PPO), using two Critic networks $V_{\phi_1}(S_t)$ and $V_{\phi_2}(\tilde{S}_t)$.

4 Interleaved Generalized Advantage Estimation (GAE)

We construct interleaved TD-errors for step t to capture credit assignment across both stages without discounting within the same time step t:

4.1 Intra-timestep TD-Errors

$$\delta_t^{(1)} = R_t^p + V_{\phi_2}(\tilde{S}_t) - V_{\phi_1}(S_t) \quad \text{(Pricing TD-error)} \tag{4}$$

$$\delta_t^{(2)} = R_t^d + \gamma V_{\phi_1}(S_{t+1}) - V_{\phi_2}(\tilde{S}_t) \quad \text{(Dispatching TD-error)} \tag{5}$$

4.2 Interleaved Advantages

With GAE parameter $\lambda \in [0, 1]$, the advantage estimates are recursively computed backward from $t = T$ down to $t = 1$:

$$\hat{A}_{2,t} = \delta_t^{(2)} + \gamma\lambda\hat{A}_{1,t+1} \quad \text{(Stage 2 Dispatching Advantage)} \tag{6}$$

$$\hat{A}_{1,t} = \delta_t^{(1)} + \lambda\hat{A}_{2,t} \quad \text{(Stage 1 Pricing Advantage)} \tag{7}$$

Remark: $\hat{A}_{1,t}$ explicitly incorporates both the immediate pricing reward R_t^p, the current dispatching performance $\delta_t^{(2)}$, and future expectations, allowing the pricing policy to receive direct feedback on how price choices impact downstream MCV fleet operational dispatch.

5 Joint PPO Objectives and Loss Functions

5.1 Stage 1 Pricing Policy Loss

Let $r_t^{(1)}(\theta_1) = \frac{\pi_{\theta_1}(\boldsymbol{p}_t^{\boldsymbol{M}}|S_t)}{\pi_{\theta_1^{\text{old}}}(\boldsymbol{p}_t^{\boldsymbol{M}}|S_t)}$ be the probability ratio.

$$L_1^{\text{CLIP}}(\theta_1) = \hat{\mathbb{E}}_t\left[\min\left(r_t^{(1)}(\theta_1)\hat{A}_{1,t}, \text{clip}(r_t^{(1)}(\theta_1), 1-\epsilon, 1+\epsilon)\hat{A}_{1,t}\right)\right] \tag{8}$$

$$L_1^{\text{VF}}(\phi_1) = \hat{\mathbb{E}}_t\left[\left(V_{\phi_1}(S_t) - (\hat{A}_{1,t} + V_{\phi_1^{\text{old}}}(S_t))\right)^2\right] \tag{9}$$

5.2 Stage 2 Dispatching Policy Loss

Let $r_t^{(2)}(\theta_2) = \frac{\pi_{\theta_2}(\mathbf{a}_t|\tilde{S}_t)}{\pi_{\theta_2^{\text{old}}}(\mathbf{a}_t|\tilde{S}_t)} = \prod_{m=1}^{n} \frac{\pi_{\theta_2}^m(a_t^m|\tilde{S}_t)}{\pi_{\theta_2^{\text{old}}}^m(a_t^m|\tilde{S}_t)}$ be the multi-discrete joint ratio.

$$L_2^{\text{CLIP}}(\theta_2) = \hat{\mathbb{E}}_t\left[\min\left(r_t^{(2)}(\theta_2)\hat{A}_{2,t}, \text{clip}(r_t^{(2)}(\theta_2), 1-\epsilon, 1+\epsilon)\hat{A}_{2,t}\right) + \beta\sum_{m=1}^{n}\mathcal{H}(\pi_{\theta_2}^m)\right] \tag{10}$$

$$L_2^{\text{VF}}(\phi_2) = \hat{\mathbb{E}}_t\left[\left(V_{\phi_2}(\tilde{S}_t) - (\hat{A}_{2,t} + V_{\phi_2^{\text{old}}}(\tilde{S}_t))\right)^2\right] \tag{11}$$

where $\mathcal{H}$ is the policy entropy bonus for encouraging MCV routing exploration.

6 Algorithm

Algorithm 1 End-to-End Joint PPO for Dynamic Pricing and MCV Dispatching

Initialize networks $\pi_{\theta_1}, V_{\phi_1}$ (Pricing) and $\pi_{\theta_2}, V_{\phi_2}$ (Dispatching)
for iteration $= 1, 2, \ldots, M$ **do**
  Initialize replay buffer $\mathcal{B} \leftarrow \emptyset$
  for episode $= 1, \ldots, N$ **do**
    Observe initial state $S_1 = (\{l_{m1}, e_{m1}\}_{m=1}^{n}, \boldsymbol{\lambda_1}, \boldsymbol{p_1^F})$
    for $t = 1, \ldots, T$ **do**
      Sample pricing action $\boldsymbol{p_t^M} \sim \pi_{\theta_1}(\cdot|S_t)$, receive immediate reward R_t^p
      Realize EV demand $\xi \sim P(\xi|\boldsymbol{p_t^M})$ and calculate $\boldsymbol{D_t} = \mathcal{D}(\boldsymbol{\lambda_t}, \boldsymbol{p_t^M}, \boldsymbol{p_t^F}; \xi)$
      Formulate intermediate state $\tilde{S}_t = (\{l_{mt}, e_{mt}\}_{m=1}^{n}, \boldsymbol{p_t^M}, \boldsymbol{p_t^F}, \boldsymbol{D_t})$
      Forward $\tilde{S}_t$ through π_{θ_2} and sample joint dispatch action $\mathbf{a}_t = [a_t^1, \ldots, a_t^n]$
      Execute $\mathbf{a}_t$ on MCV fleet; receive profit R_t^d and observe next macro state S_{t+1}
      Store transition $(S_t, \boldsymbol{p_t^M}, R_t^p, \tilde{S}_t, \mathbf{a}_t, R_t^d, S_{t+1})$ in $\mathcal{B}$
    end for
  end for
  Compute Interleaved GAE Advantages Backward
  Set $\hat{A}_{1,T+1} = 0$
  for $t = T, T-1, \ldots, 1$ **do**
    Compute $\delta_t^{(2)} = R_t^d + \gamma V_{\phi_1}(S_{t+1}) - V_{\phi_2}(\tilde{S}_t)$
    Compute $\hat{A}_{2,t} = \delta_t^{(2)} + \gamma\lambda\hat{A}_{1,t+1}$
    Compute $\delta_t^{(1)} = R_t^p + V_{\phi_2}(\tilde{S}_t) - V_{\phi_1}(S_t)$
    Compute $\hat{A}_{1,t} = \delta_t^{(1)} + \lambda\hat{A}_{2,t}$
  end for
  PPO Mini-Batch Updates
  for epoch $= 1, \ldots, K$ **do**
    Update Pricing Actor θ_1 by maximizing $L_1^{\mathrm{CLIP}}(\theta_1)$
    Update Pricing Critic ϕ_1 by minimizing $L_1^{\mathrm{VF}}(\phi_1)$
    Update Dispatching Actor θ_2 by maximizing $L_2^{\mathrm{CLIP}}(\theta_2)$
    Update Dispatching Critic ϕ_2 by minimizing $L_2^{\mathrm{VF}}(\phi_2)$
  end for
  Synchronize old policies $\theta_1^{\mathrm{old}} \leftarrow \theta_1$, $\theta_2^{\mathrm{old}} \leftarrow \theta_2$
end for=0